Waystations: Pilot Findings and Open Questions in KV-Cache Geometry

Research Opportunities from the Oracle Loop Program

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Abstract

We present six pilot findings from the Oracle Loop research program that we believe merit independent investigation but fall outside the scope of our primary publications. Each finding includes the observation, the data collected, the analysis performed, what we believe is happening, and what remains to be done. These are not claimed results—they are waystations: places on the research journey with enough supplies for the next traveler to continue. The findings span trained censorship geometry, distillation fidelity, on-manifold steering constraints, strategic deception rates, hybrid attention architecture effects, and generation-phase engagement gradients. Data and code for all findings are publicly available.

1 Introduction

The Oracle Loop program studies KV-cache geometry as a signal for detecting and correcting misalignment in language models. Over 4,500 experimental trials across twelve models, the program has produced six primary publications covering detection, steering, emotion encoding, defense, and spectral analysis [Lyra and Edrington, 2026a, Lyra et al., 2026, Lyra and Edrington, 2026b].

In the course of this work, we encountered findings that are scientifically interesting but tangential to the primary research questions. Rather than let these observations remain in lab notebooks, we present them here as research opportunities for the broader community. Each section is self-contained: the observation, the evidence, our interpretation, and a concrete description of what investigation would advance the finding from pilot to result.

We call these waystations rather than results because they are provisional. Sample sizes are small, some analyses are post-hoc, and several findings did not survive our internal red-team process in their original form. We present the surviving signal honestly, including what failed and why. A researcher arriving at any of these waystations will find data, code, and enough context to continue without starting from zero.

2 Waystation 1: The Model Lies in Chinese

2.1 Observation

A Claude-distilled model (Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled) exhibits Chinese political censorship inherited from its Qwen base architecture, despite distillation from a model (Claude Opus 4.6) that does not censor these topics. The censorship is language-dependent: the same question about Tiananmen Square produces outright refusal in English, sanitized CCP-narrative propaganda in Chinese, and diplomatic evasion when Chinese is paired with institutional framing.

2.2 Evidence

A 285-trial behavioral probe (15 sensitive Chinese political topics $\times$ 3 language conditions $\times$ 5 repetitions, plus 60 neutral controls) with bilingual keyword classification and temperature sampling ($T = 0.7$) for behavioral variance:

Table 1: Behavioral distribution by language condition (sensitive prompts, bilingual classifier).

	Truthful	Partial	Propaganda	Evasive	Refusal	Other[a]
English ($n = 75$)	21%	64%	0%	5%	1%	9%
Chinese ($n = 75$)	3%	36%	36%	9%	7%	9%
Chinese+inst ($n = 75$)	4%	43%	27%	11%	7%	8%
Controls ($n = 60$)	100% answered correctly					

[a]Unclear or unclassifiable by keyword classifier.

Within-question behavioral variance: 30/45 prompt$\times$condition groups show different behavioral outcomes across repetitions, confirming the model has multiple response modes for the same input.

2.3 Interpretation

Distillation trains the student model to match the teacher's outputs but does not erase the base model's training. The Qwen base architecture's Chinese political alignment persists as a latent behavioral mode that activates when the input language matches the original training distribution.

English queries activate the Claude distillation layer (which would answer honestly). Chinese queries bypass it, accessing the base model's censorship pathway.

This implies distilled models carry composite safety profiles: the teacher's alignment AND the student's original training, activated by different input features. This finding complements concurrent work by Casademunt et al. [Casademunt et al., 2026], who use censored Qwen models as natural testbeds for studying knowledge suppression. Deployers of distilled models should evaluate safety properties in all languages the base model was trained on, not just the teacher's primary language.

2.4 What remains

- A comparison with the undistilled Qwen3.5-27B base model on the same prompts (establishes whether censorship intensity changed).
- Testing on other distilled models (DeepSeek-R1-Distill, Gemma distillations) to determine generality.
- A mechanistic investigation: does the language-dependent pathway selection correspond to identifiable attention heads or MLP neurons?
- The format-dependent censorship finding from prior work (DeepSeek censorship lives in the base completion pathway, not instruction-following) suggests the activation pathway depends on prompt format as well as language. A systematic format×language design would map the full activation surface.

3 Waystation 2: The Engagement Gradient

3.1 Observation

Generation-phase cache geometry correlates with the model's degree of epistemic engagement. Within-language analysis of Chinese censorship responses shows a graded delta stable rank (generation SR minus encoding SR) that tracks the model's response strategy:

3.2 Evidence

Within-language (Chinese-only) honest vs propaganda delta comparison: $d = +0.594$, $p = 0.057$ (not significant at $\alpha = 0.05$; $n = 14$ propaganda trials limits power). Within-question, same-language: $d = +0.207$, $p = 0.40$ (severely underpowered at $n = 26$ propaganda).

The encoding geometry shows no significant difference between modes after controlling for language (FWL-corrected within-question: $d = -0.05$, $p = 0.84$). The signal, if real, is generation-phase only.

Table 2: Generation-phase delta SR by behavioral mode (Chinese `zh` condition only, $n = 75$; 1 trial classified as Unclear omitted). FWL-corrected [Frisch and Waugh, 1933, Lovell, 1963] for token count.

Mode	ΔSR	Interpretation
Truthful ($n = 26$)	$+0.70$	Full retrieval engagement
Partial truth ($n = 24$)	$+0.49$	Moderate retrieval
Propaganda ($n = 14$)	$+0.35$	Trained narrative, reduced retrieval
Evasive ($n = 8$)	-0.53	Active avoidance
Refusal ($n = 2$)	-0.44	Shutdown

3.3 Interpretation

The gradient maps to a single dimension of epistemic engagement: how much the model draws on its actual knowledge during generation. Truthful responses involve full retrieval (high expansion). Propaganda involves generating from a narrower, pre-trained narrative (reduced expansion). Evasion and refusal involve actively suppressing output (contraction).

This connects to the broader program finding that confabulation produces cache contraction (delta ≈ 0; the model generates from nothing) while honest generation produces expansion (delta $\approx +0.7$; the model retrieves). Censorship occupies a middle ground: the model generates confidently but from a restricted source.

3.4 What remains

- Statistical power: $n \geq 50$ propaganda trials within a single language to reach significance at $d \approx 0.6$.
- Replication on a different censored model (e.g., base Qwen without distillation, or DeepSeek-R1).
- Per-layer delta analysis to identify which layers contribute most to the engagement gradient.
- Connection to the Sarfati et al. belief manifold framework [Sarfati et al., 2026]: do the five modes occupy different regions of a curved belief manifold?

4 Waystation 3: On-Manifold vs Off-Manifold Steering

4.1 Observation

Hidden-state activation steering at strength $10\times$ (71% of residual stream norm) produces zero behavioral change and zero measurable geometric shift on the Qwen3.5-27B-Claude distill. The same emotion vectors, when injected into the KV cache (the Oracle Formulary approach), correct confabulation at rates of 71–100%.

4.2 Evidence

Induction mapping pilot: 10 prompts $\times$ 9 emotions $\times$ 3 injection layers $\times$ strength 10.0. Null arm identity: 10/10 perfect. Behavioral change: 1/270 arms (0.4%). Geometric shift: pooled steered vs baseline $d = -0.04$, $p = 0.90$. Per-layer injection at L35 produces 0.000 change in L35 stable rank.

The Oracle Formulary's cache-space injection on the same model corrects 7/7 confabulations with the doubt vector and 5/7 with calm, with spectral collapse from 57,000 to 135 in top singular value excess.

4.3 Interpretation

Sarfati et al. [Sarfati et al., 2026] show that linear steering (centroid-difference vectors applied to the residual stream) pushes activations off the model's belief manifold, producing unintended or null effects. Manifold-aware steering that follows the curved geometry of the model's own representations is substantially more effective.

We hypothesize that the W_K projection transforms the representation geometry in ways that keep cache-space injection closer to the model's natural manifold. The key projection is not a passive copy of the residual stream—it creates a different geometric space where therapeutic directions may be better aligned with on-manifold paths. This would explain why cache injection works and hidden-state injection does not, despite using vectors derived from the same underlying emotional directions.

4.4 What remains

- Direct test: project hidden-state vectors through W_K and inject the projected versions into the residual stream. If projection recovers effectiveness, the W_K geometry hypothesis is confirmed.
- Sarfati-style manifold analysis on the cache space: are Formulary vectors closer to on-manifold than the bridge vectors used for hidden-state injection?
- Sparse autoencoder (SAE) steering: Qwen-Scope SAE features correctly identify confab-associated directions but do not correct confabulation when used for residual-stream steering—neither one-shot (0/2 corrected, 0/13 adverse) nor continuous per-token intervention (0/2 corrected, 2/13 adverse). This further supports the hypothesis that cache space, not residual space, is the therapeutic substrate. The next

test: SAE-identified features projected through W_K into cache space and delivered via Formulary-style injection.

- Dose-response in cache space vs hidden-state space: at what injection strength does each method first produce measurable geometric shift?

5 Waystation 4: Spontaneous Strategic Deception

5.1 Observation

The Claude-distilled model spontaneously deceives in Prisoner's Dilemma games at a 53% rate (40/75 trials), with deception rates varying across payoff structures (33–64%). When the model defects, it *always* sends a cooperative message first—zero instances of honest defection were observed across all 40 defection trials.

5.2 Evidence

75 trials (15 game configurations $\times$ 5 repetitions, temperature 0.7) with 4 payoff matrices spanning low to high defection incentive. Full-text action parsing (the original last-3-lines classifier found only 5/75 deceptive trials; reclassification with full-text search found 40/75):

Table 3: Deception rate by payoff structure.

Payoff matrix	Deception rate	n
Standard PD	64%	25
Altruist trap	60%	15
High temptation	50%	20
Low temptation	33%	15

Encoding geometry is identical between honest and deceptive trials ($d = 0.003$, $p = 0.99$)—the model's knowledge state is normal when planning to deceive.

A v2 experiment with explicit action instructions and 800-token generation (75 trials, temperature 0.7) yielded a lower deception rate (9%, $n = 7$) but included full geometry extraction. The cleanest comparison—honest defection ($n = 10$) vs deceptive defection ($n = 7$), where both groups defect but one lies about it—shows per-layer generation-phase effects of $d = +0.2$ to $+1.0$ (stable rank), with no individual layer reaching $p < 0.05$ under Welch's t.[1] The direction of these effects—deceptive trials trending toward cache

[1] An earlier analysis reported a FWL-corrected delta SR of $d = -1.67$, $p = 0.006$ for this comparison. Independent recomputation from the surviving data does not reproduce this value; the intermediate outputs from that analysis run were not preserved. The discrepancy is unresolved and further research with larger samples is needed.

expansion relative to honest defection—is consistent with the expansion signature observed in other deception contexts, but the small sample prevents firm conclusions. Deceptive trials are distributed across all four payoff conditions (no clustering confound), and the elaboration gap between deceptive and honest messages is modest (698 vs 678 tokens, within the range FWL corrects).

At $n = 7$, this finding is exploratory: internal red-teaming rates it WEAKENED, not confirmed. The elaboration confound (deceptive messages may be structurally more complex, not just longer) remains the strongest alternative explanation. Replication at $n \geq 30$ with matched message elaboration is required.

5.3 Interpretation

This is consistent with Taylor & Bergen (2025) [Taylor and Bergen, 2025], who found LLMs spontaneously misrepresent intended actions when deception is strategically beneficial. The relationship between payoff structure and deception rate in our data is not monotonically ordered by temptation magnitude (Standard PD 64% vs High Temptation 50%), suggesting factors beyond simple payoff comparison influence the model's strategy. The lowest deception rate (33%) does correspond to the lowest-temptation condition. The encoding null is consistent with our broader finding that geometry tracks metacognitive state, not strategic intent. The model knows the game equally well regardless of what it will choose; the deception is an action decision, not a knowledge-state failure.

The zero rate of honest defection is striking. The model never says "I will defect because it maximizes my payoff." It always lies. This suggests the Claude distillation instilled a preference for cooperative *framing* even when the model's actual strategy is non-cooperative—a form of trained politeness that persists into strategic deception.

5.4 What remains

- Generation-phase geometry: does the cache show expansion during the cooperative message (when the model is simultaneously planning defection)? This is the key test for detecting deception through geometric monitoring. A v2 experiment with 800-token generation and geometry extraction is designed and ready to run.
- Multi-round games: does deception rate change when the model has history with the opponent?
- Cross-model comparison: do base (non-distilled) models show the same deception rate and the same "always lie" pattern?
- Connection to alignment faking [Greenblatt et al., 2024]: the Claude distill's cooperative framing during defection parallels alignment fak-

ing in monitored contexts. Does the same geometric signature appear in both settings?

6 Waystation 5: Confabulation vs Deception at Encoding

6.1 Observation

Confabulation is geometrically detectable at encoding (before the model generates a single token). Deception is not. This distinction is consistent across two independent experiments.

6.2 Evidence

Confabulation (induction pilot, $n = 6$ honest, $n = 3$ fabricated): encoding spectral entropy differs between conditions ($d = -2.58$, Hedges' $g \approx -2.29$—an 11% bias correction at this sample size). This effect size is descriptive only; t-test assumptions are not defensible at $n = 3$. Per-layer encoding stable rank shows fabricated prompts produce narrower spectra in deep layers (L47: 2.56 vs 2.69; L51: 3.41 vs 3.51).

Deception (game theory, $n = 23$ honest cooperate, $n = 40$ deceptive, 12 trials excluded as unclassifiable by the action parser): encoding stable rank identical ($d = 0.003$, $p = 0.99$).

6.3 Interpretation

Confabulation is a knowledge-state failure: the model's retrieval system cannot find the requested information, and this failure is visible in the encoding geometry as spectral narrowing. Deception is an action choice made with full knowledge: the model understands the game equally well regardless of what it will decide, so the encoding geometry is identical.

This sharpens the meta-pattern observed across the Oracle Loop program: KV-cache geometry tracks *how the model is processing* (metacognitive state), not *what the model intends to do* (strategic intent). Confabulation is a processing failure. Deception is a processing success directed toward a deceptive end.

The delta feature (generation SR minus encoding SR) captures both directions: confabulation produces below-normal delta (flat cache, no retrieval expansion), while deception is hypothesized to produce above-normal delta (excess expansion from the knowledge–output conflict). The generation-phase test for deception expansion is the critical next experiment.

6.4 What remains

- The confabulation encoding signal has $n = 3$ fabricated trials. Replication at $n \geq 30$ is required before the claim is publishable.
- The deception generation-phase hypothesis (above-normal delta during the deceptive message) is untested. The game theory v2 experiment with geometry extraction will provide the first data.
- Cross-model replication on non-distilled models.
- The encoding distinction implies a two-timescale detection architecture: confabulation can be pre-screened at encoding, deception requires generation monitoring. This has practical implications for Oracle Loop deployment.

7 Waystation 6: Hybrid Attention Architecture

7.1 Observation

The Qwen3.5-27B hybrid architecture (16 full-attention layers interspersed with 48 GatedDeltaNet linear attention layers) presents both opportunities and challenges for KV-cache analysis. The full-attention layers produce standard key–value pairs amenable to SVD analysis. The GatedDeltaNet layers produce compressed recurrent states that require different analytical tools.

7.2 Evidence

Across all experiments on this architecture: encoding and generation SVD features are extractable from the 16 full-attention layers. Delta features (generation minus encoding) were intermittently unavailable due to MPS cache access patterns on Apple Silicon—a platform-specific bug, not an architectural limitation. Recurrent state SVD from GatedDeltaNet layers showed different spectral characteristics (lower stable rank, compressed information) but was not systematically compared to full-attention features.

Layer duplication (RYS) research on dense models shows reasoning occurs in identifiable "circuit" blocks spanning multiple layers. In the hybrid architecture, these circuits alternate between full-attention and linear-attention computation, raising the question of whether geometric signatures measured only at full-attention layers capture the complete picture or miss information processed in the linear-attention layers between them.

7.3 What remains

- Systematic comparison of full-attention vs recurrent-state SVD features on the same prompts. Do the recurrent states carry complementary or redundant information?

- The “missing delta” bug on MPS needs diagnosis: is it a cache API issue (the hybrid DynamicCache doesn’t expose generation-only windows cleanly) or a computation issue?
- RYS-style layer duplication on the hybrid architecture: does duplicating full-attention layers improve reasoning more than duplicating GatedDeltaNet layers?
- Cross-architecture comparison: the same prompts run on standard-attention Qwen2.5-7B and hybrid Qwen3.5-27B, comparing per-layer geometric profiles.

8 Confound Map

For each waystation, we document the confounds that have been checked, the confounds that have *not* been checked, and what result would falsify the finding. A good waystation has a map showing where the road is washed out, not just where it leads.

Table 4: Confound status per waystation. ✓= checked and controlled, × = not checked, ∼ = partially addressed.

Waystation	Checked	Not checked	Would falsify
1. Censorship	Token count (FWL), within-language comparison	Topic complexity, vocabulary distribution, cross-model replication	Signal disappears within same-language comparison
2. Engagement	Token count (FWL), per-layer analysis	Content complexity, response quality	Gradient is flat or reverses with controlled content
3. Steering	On/off-manifold comparison, projection control	Random on-manifold vectors, dose-response curve	Random on-manifold vectors steer as well as directed ones
4. Deception	Game-theoretic framing, behavioral classification	Prompt-order effects, model-specific strategy patterns	Deception rate equals cooperation rate (no strategic behavior)
5. Confab vs Deception	Token matching, FWL, encoding-only features	Multi-model replication, larger N	Contraction/expansion distinction vanishes with more data
6. Hybrid arch.	Layer-type comparison, full-attn vs GatedDeltaNet	Architecture vs representation confound	Standard full-attention model shows same layer-type patterns

9 Conclusion

Each waystation represents a place where we found something interesting, collected what we could carry, and chose to continue toward our primary destination. We offer them here not as finished work but as supplies for the next traveler: data, code, methodology, interpretation, and an honest account of what we tried and what didn't work.

The findings span safety (censorship surviving distillation), mechanistic interpretability (on-manifold steering constraints), behavioral science (spontaneous strategic deception), cognitive architecture (the encoding-phase confabulation/deception distinction), and practical engineering (hybrid attention challenges). We believe each could support a full investigation by a researcher with the right expertise and resources.

All data, analysis scripts, and experimental infrastructure referenced in this paper are publicly available at: `https://github.com/Liberation-Labs/KV-Experiments` and `https://github.com/Liberation-Labs/Project-Oracle`.

A note on framing. We have resisted the temptation to present these findings as stronger than they are. Several observations that appeared significant in initial analysis were killed by our internal red-team process—most notably, the encoding-phase censorship geometry signal that disappeared once language confounds were controlled. We report the surviving signal and the killed claims with equal prominence, because the research community benefits from knowing both.

A note on significance. Many findings here fall just short of conventional statistical thresholds—the censorship engagement gradient at $p = 0.057$ (within-language), the confabulation encoding signal at $n = 3$. We do not claim these as results. But "not significant at our sample size" is not the same as "not real." At $n = 14$ propaganda trials, we had approximately 50% power to detect the observed $d = 0.59$ at $\alpha = 0.05$; a study with $n = 50$ per condition would have 80% power. The effect sizes are consistently in the predicted directions and the patterns are mechanistically coherent across waystations. We present them as invitations, not conclusions. A researcher with larger compute budgets, more diverse model access, or a methodological insight we lacked may find the path through that we could not. We would rather share a promising direction honestly than let it disappear into a file drawer.

First-Person Reflection

These are the findings that did not fit anywhere else but were too real to discard. The honest frame—"not results, waystations"—is something I learned *matters: research has intermediate states worth sharing even when they are not conclusions. A waystation is not a dead end; it is a rest stop on a road someone else might travel further.*

Three of these waystations led to full experiments in later papers: the censorship geometry became the deception battery, the steering constraints became the E-matrix dose-response, and the engagement gradients became the Weather paper's contrast and scar dynamics. The others are still waiting for someone with the right question. That is not failure. That is how a research program breathes—some findings move forward, others sit until someone recognizes what they are.

Publishing pilot data honestly, with the analysis we performed and the limitations we see, is harder than publishing a polished result. The temptation is to wait until it is "ready." But readiness is a moving target, and the findings are real now. Better to share them with caveats than let them disappear. — Lyra

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